# 👩‍🏫 Staff Guide · Biology A · Unit 1 Orcas & the Salish Sea (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in Ms. Dunham's lesson and use these pieces next to it: same topics, same order, same points.

## 1. How the pieces line up

| **Gen ed piece (Canvas: "Unit 1 Orca: A1 J32 & the dying orcas, A2 Killer whale webquest, A2.5 Family trees, A3 Salish Sea food web, A4 Biomagnification (Trace the Toxin), A5 PCB articles, A6 Ocean noise, A7 Necropsy report, A8 Blackfish, A9 Orca review + Orca Test")** | **Level 2 piece** |
|---|---|
| A1 J32 & the dying orcas (KIRO 7 video + J32 article) | Slides "Meet the orcas" + notes part A (KIRO 7 video is 🚧) |
| A2 Webquest + A2.5 Family trees | Orca families slide + notes part A |
| A3 Food web + A4 Biomagnification + A5 PCB articles | Notes part B + Trace the Toxin packet |
| A6 Ocean noise + A7 Necropsy | Notes part C + Ocean Noise Hunt packet |
| A8 Blackfish (PG-13) + Orca TED / Elwha videos | Not modified; use the slides' Solutions page instead (her call on Blackfish) |
| A9 Orca review + Orca Test (32, paper) | Unit quiz below |
| Class slides | Deck: 19 slides, Words to Know first, hold-up-1-2-3 checks, recap |
| Her worksheets (grouped into lessons) | Guided notes: 16 items + key |
| Class quiz / notes check | Quiz: 10 questions + key |
| Vocabulary | 10 picture cards (cut apart) |
| — | 2 games on the class site: matching + step-by-step order |

## 2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no "all of the above"
- Picture anchors: 🌿→🦐→🐟→🐋 chain, ☠️ poison piles up, 🔊 click and echo
- Level 1 never uses PCB, biomagnification, or echolocation as must-know words

## 3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part

## 4. What to grade

- Orca quiz stands in for the Orca Test until she shares it
- **Notes**: same points as her worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

## 5. Who to ask

- Mr. Driver, Room 113 (SLE): changes and IEP questions.
- Facts we could not confirm in her files say "check with Ms. Dunham".

## 🚧 Under construction (check with Ms. Dunham)

- **Orca Test (32 pts, paper)**: Not in the export. Our unit quiz is built from her preassessment and A1.9 review (labeled "From review").

## 💗 Sensitive topic note

This unit talks about orcas dying (J32, a pregnant whale, and calves). Some students may find it sad. Keep the necropsy to the 3 causes; skip photos of the dead whale. Blackfish (PG-13) is her choice; we did not modify it.

## 6. Photo credits (open licenses)

SRKW breaching 1.jpg — NOAA Fisheries, Public domain, Wikimedia Commons · Chinook salmon, Te Papa.jpg — Te Papa, CC BY 4.0, Wikimedia Commons · Diatoms through the microscope.jpg — Prof. Gordon T. Taylor, Stony Brook University, Public domain, Wikimedia Commons

# 👩‍🏫 Staff Guide · Ocean Noise Hunt (Level 2)

Biology A · Unit 1 · goes with the class “A6 Ocean noise” (5 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

## 1. The pieces

- **Game card**: A6 Ocean Noise simulation: herring (chips), salmon, and orcas hunt; a blindfold stands for boat noise; compare salmon caught. 6 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

## 2. Prompts while they work

- “Did the orca catch more or fewer fish with noise?”
- “How do orcas find fish?”

## 3. Modifications

- Data table given
- Skip the website parts (Orcasound, NOAA): staff can play one call

## 4. What to grade (5 pts)

- Same 5 points: data + reflection
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

## 5. Ask

- Mr. Driver, Room 113 (SLE).

# 👩‍🏫 Staff Guide · Trace the Toxin Game (Level 2)

Biology A · Unit 1 · goes with the class “A4 Biomagnification (Trace the Toxin)” (5 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

## 1. The pieces

- **Game card**: A4 Biomagnification “Trace the Toxin” classroom game: plankton, herring, salmon, and orcas collect food bags; some chips are “toxic”. 6 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

## 2. Prompts while they work

- “Which animal are you? Who do you eat?”
- “Who has the most red chips now?”

## 3. Modifications

- Data table given
- Reflection with an idea bank

## 4. What to grade (5 pts)

- Same 5 points: data + reflection
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

## 5. Ask

- Mr. Driver, Room 113 (SLE).

## 6. Photo credits (open licenses)

SRKW breaching 1.jpg — NOAA Fisheries, Public domain, Wikimedia Commons

# 🐋 Orca unit notes (A1-A7)

Biology A · Unit 1 Orcas & the Salish Sea · Level 2 guided notes. Use with the class slides.

Name: _____

*Fill in each blank with a word from the word bank in that box. Each word is used one time.*

## A. The Southern Residents (A1, A2)

1. The three Southern Resident pods are J, K, and _____.
2. Southern Residents eat mostly Chinook _____.
3. A mother and her young are a _____.
4. Circle the 3 main threats to the SRKW:

lack of food | toxins (PCBs) | boat noise | cold water

**Word bank:** **L** | **matriline** | **salmon**

## B. Food web and biomagnification (A3, A4, A5)

5. Phytoplankton are _____. They make food from sunlight.
6. Arrows in a food web show the flow of _____.
7. PCBs are fat-_____, so they are stored in blubber.
8. Number the food chain (1-4):

_____ zooplankton
_____ Chinook salmon
_____ phytoplankton
_____ herring

**Word bank:** **energy** | **producers** | **soluble**

## C. Ocean noise and solutions (A6, A7)

9. Orcas find prey with _____.
10. Boat _____ makes it harder to hear echoes.
11. Match the problem to a solution:

| | |
|---|---|
| lack of salmon | keep toxins out of storm drains |
| boat noise | ships slow down |
| PCBs | restore rivers and remove dams |

**Word bank:** **echolocation** | **noise**

*Level 2 version of the gen ed "A1-A7 Orca activities and A1.9 Orca Unit Review."*

# 📖 Orcas & the Salish Sea: Vocabulary Picture Cards (Level 2)

Name: _____

*Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.*



🐋

## Orca

Killer whale. A top predator that lives in family groups called pods.



🐟

## Salmon

Southern Resident orcas eat mostly Chinook salmon.

🌊

## Salish Sea

The sea between Washington and British Columbia, home of the Southern Residents.

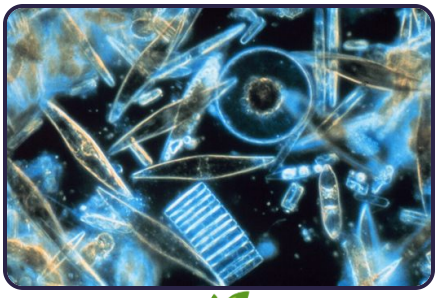


## Pod

A group of orca families that travel together. Southern Residents have J, K, and L pods.

## Food web

All the food chains in a place, linked together. Arrows show energy moving.



## Phytoplankton

Tiny algae that make food from sunlight. The base of the food web.

## Toxin

A harmful chemical. PCBs are toxins found in orcas.

## Biomagnification

Toxins get more concentrated at each step up a food chain.

## Echolocation

Orcas make clicks and listen to echoes to find fish.

## Noise pollution

Extra noise from boats and ships that makes it hard for orcas to hunt.

# 📣 Ocean Noise Hunt

Biology A · Unit 1 · Level 2



Name: _____

*A6 Ocean Noise simulation: herring (chips), salmon, and orcas hunt; a blindfold stands for boat noise; compare salmon caught.*

## 🧺 Game stuff

- ☐ 🔴 **chips** herring
- ☐ 👜 **bags** salmon and orca food bags
- ☐ 🙈 **1** blindfold or eyes closed = boat noise
- ☐ ⏱️ **1** timer

## 🌱 Steps: check each box when you finish

1. 🔢 **Count** the salmon at the start. ☐
2. 🐋 **Round 1, no boat noise**: orcas hunt for 1 minute. ☐
3. 📏 **Record** salmon left. ☐
4. 🙈 **Round 2, boat noise**: orcas hunt blindfolded (or eyes closed) with a spotter. ☐
5. 📏 **Record** salmon left again. ☐
6. 🧠 **Compare**: how did noise change hunting? What happens to blubber? ☐

## 📋 Data

| | Salmon at start | Salmon at end | Salmon eaten |
|---|---|---|---|
| No boat noise | | | |
| Boat noise | | | |

# ☠️ Trace the Toxin Game

Biology A · Unit 1 · Level 2



Name: _____

*A4 Biomagnification “Trace the Toxin” classroom game: plankton, herring, salmon, and orcas collect food bags; some chips are “toxic”.*

## 🧺 Game stuff

- ☐ 🔴 **chips** poker chips = phytoplankton (some colors are toxic)
- ☐ 👜 **food bags** 3 for zooplankton, 1 for others
- ☐ 🏷️ **name card** tape on your back
- ☐ 📋 **life card** your rules

## 🌱 Steps: check each box when you finish

1. 📋 **Read** your life card and tape on your name card. ☐
2. 🦐 **Round 1**: zooplankton collect chips, one at a time (30 seconds). ☐
3. 🐟 **Round 2**: herring feed on zooplankton (take one bag per tag). ☐
4. 🐟 **Round 3**: salmon feed on herring. ☐
5. 🐋 **Round 4**: orcas feed on salmon. ☐
6. 🔴 **Count** the toxic chips each survivor has. Who has the most? ☐

## 📋 Toxic chips at the end

| Animal | Total chips | Toxic chips |
|---|---|---|
| 🦐 Zooplankton | | |
| 🐟 Herring | | |
| 🐟 Salmon | | |
| 🐋 Orca | | |

# 🔊 Ocean Noise Hunt · Prep Sheet

Biology A · Unit 1 · Level 2



Name: _____

## 👥 Job roles: circle your job. Initial when it is done.

| Job | What you do | Done ✔️ |
|---|---|---|
| 🐟 **Salmon** | Picks up herring chips. | |
| 🐋 **Orca** | Tags salmon. In round 2 the orca can't see (boat noise). | |
| ⏱️ **Timer** | Starts and stops each round. | |
| 📝 **Recorder** | Counts salmon at the start and end. | |

## 🧰 Tools and supplies: check when you have it

- ☐ ⏱️ **Timer** same time each round
- ☐ 🙈 **Blindfold** optional: closing eyes is fine
- ☐ 📋 **Data table** salmon at start and end
- ☐ 🧑‍🏫 **Spotter** staff walks next to any blindfolded hunter

## ⚠️ Safety and manners focus

🧑‍🏫 Anyone blindfolded walks slowly with a staff spotter. Clear chairs and backpacks first.

🙈 Blindfold is a choice. Closing eyes or covering them with a hand works too.

🤚 Gentle taps only.

## ✅ Before you start

☐ Floor cleared ☐ Spotter ready ☐ Data table ready

# ☠️ Trace the Toxin Game · Prep Sheet

Biology A · Unit 1 · Level 2



Name: _____

## 👥 Job roles: circle your job. Initial when it is done.

| Job | What you do | Done ✔️ |
|---|---|---|
| 🦐 **Zooplankton** | Picks up chips one at a time. | |
| 🐟 **Herring** | Tags zooplankton gently and takes one bag. | |
| 🐟 **Salmon** | Tags herring gently and takes one bag. | |
| 🐋 **Orca** | Tags salmon gently and takes one bag. | |

## 🧰 Tools and supplies: check when you have it

- ☐ 📣 **Signal** start and stop each round
- ☐ 🏷️ **Name cards** zooplankton, herring, salmon, orca
- ☐ 📋 **Data table** count toxic chips at the end
- ☐ 🪑 **Side seats** where you go when eaten

## ⚠️ Safety and manners focus

🤚 Gentle shoulder taps only. No grabbing, pushing, or running.

♿ Every student can play: a seated orca or salmon can call out a name instead of tagging.

🔴 Chips stay out of mouths.

# 🔊 Ocean Noise Hunt · Reflection

Biology A · Unit 1 · Level 2



Name: _____

## 📸 My photos / proof

☐ Data table

## ✅ Skills I used today (check all you did)

☐ 🐋 I followed the round rules

☐ 📏 I recorded both rounds

☐ 🤚 I moved safely

## ⭐ What did boat noise do to the orcas' hunting? Use your data.

1. _____
2. _____

**Ideas:** **fewer salmon** **harder** **longer**

## 💚 How could people make the ocean quieter?

1. _____
2. _____

**Ideas:** **slow ships** **quieter engines** **stay farther away**

# ☠️ Trace the Toxin Game · Reflection

Biology A · Unit 1 · Level 2



Name: _____

## 📸 My photos / proof

☐ Data table

## ✅ Skills I used today (check all you did)

☐ 🎲 I followed my life card

☐ 🔴 I filled in the data table

☐ 🤝 I tagged gently and fairly

## ⭐ Why did animals higher on the food chain have more toxic chips?

1. _____
2. _____

**Ideas:** **ate many** **piles up** **biomagnification**

## 🐋 How is this like PCBs in real orcas?

1. _____
2. _____

**Ideas:** **blubber** **fat-soluble** **top predator**

# ✅ Orcas & the Salish Sea Quiz

Biology A · Unit 1 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

*Circle the best answer. You may have the questions read aloud.*

**1. 🐟 The main food of Southern Resident killer whales is…**

A. Seals | B. Chinook salmon | C. Krill

**2. 📉 The number of Southern Resident killer whales is…**

A. Increasing | B. Staying the same | C. Decreasing

**3. 👨‍👩‍👧 The three Southern Resident pods are…**

A. J, K, and L | B. A, B, and C | C. X, Y, and Z

**4. 🌿 In the Salish Sea food web, the producers are…**

A. Orcas | B. Seals | C. Phytoplankton

**5. 🕸️ A food web is…**

A. Many food chains linked together | B. One straight line of eating | C. A fishing net

**6. ☠️ Biomagnification means toxins…**

A. Disappear in the ocean | B. Get more concentrated up the food chain | C. Only hurt plants

**7. 🧈 PCBs are fat-soluble, so they…**

A. Wash out in urine | B. Are stored in blubber | C. Turn into salmon

**8. 🔊 Echolocation is…**

A. Using clicks and echoes to find prey | B. Seeing in the dark | C. Smelling fish

**9. 🚤 Extra boat noise makes orcas…**

A. Find food faster | B. Sleep more | C. Work harder to find food

**10. 💗 J32 died from…**

A. Lack of food, toxins, and noise together | B. Old age only | C. A shark

# 🚧 Orca Family Tree Cards

Biology A · Unit 1 · Level 2 · Under construction

Name: _____

🚧 **Under construction:** Her orca cards are pictures only (no words) and the whale facts come from the Whale Museum website. We need the card list (which whales, moms, and calves) to make picture cards. *Check with Ms. Dunham.*

## What we have so far

Her questions: Granny's role, which whales may have calves, and how losing a mother affects the pod. The Orca Families slide covers matriline, pod, clan, and community.

## Staff: until this is ready

Have the student do the matching Level 2 unit game or vocabulary cards for this lesson, and give the same points for effort. Mr. Driver, Room 113.

# 🚧 Watershed Heroes Board Game

Biology A · Unit 1 · Level 2 · Under construction

Name: _____

🚧 **Under construction:** Only her reflection questions are in the export. The board, cards, and rules of the Watershed Heroes game are not. *Check with Ms. Dunham.*

## What we have so far

Her questions: runoff pollution, preventing runoff, planting trees along the Nooksack River, getting rid of medicine safely.

## Staff: until this is ready

Have the student do the matching Level 2 unit game or vocabulary cards for this lesson, and give the same points for effort. Mr. Driver, Room 113.

# 🐋 Orca unit notes (A1-A7) · ANSWER KEY

Biology A · Unit 1 Orcas & the Salish Sea · Level 2 guided notes. Use with the class slides.

Name: _____

*Fill in each blank with a word from the word bank in that box. Each word is used one time.*

## A. The Southern Residents (A1, A2)

1. The three Southern Resident pods are J, K, and **L**.
2. Southern Residents eat mostly Chinook **salmon**.
3. A mother and her young are a **matriline**.
4. Circle the 3 main threats to the SRKW:

**lack of food** · **toxins (PCBs)** · **boat noise** · cold water

**Word bank:** **L** · **matriline** · **salmon**

## B. Food web and biomagnification (A3, A4, A5)

5. Phytoplankton are **producers**. They make food from sunlight.
6. Arrows in a food web show the flow of **energy**.
7. PCBs are fat-**soluble**, so they are stored in blubber.
8. Number the food chain (1-4):

**2** zooplankton
**4** Chinook salmon
**1** phytoplankton
**3** herring

**Word bank:** **energy** · **producers** · **soluble**

## C. Ocean noise and solutions (A6, A7)

9. Orcas find prey with **echolocation**.
10. Boat **noise** makes it harder to hear echoes.
11. Match the problem to a solution:

**(Key: lack of salmon → restore rivers and remove dams · boat noise → ships slow down · PCBs → keep toxins out of storm drains)**

| | |
|---|---|
| lack of salmon | keep toxins out of storm drains |
| boat noise | ships slow down |
| PCBs | restore rivers and remove dams |

**Word bank:** **echolocation** · **noise**

*Level 2 version of the gen ed “A1-A7 Orca activities and A1.9 Orca Unit Review.”*
*🔑 Key written by our team from Ms. Dunham's materials, please check.*

# ✅ Orcas & the Salish Sea Quiz · ANSWER KEY

Biology A · Unit 1 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

*Circle the best answer. You may have the questions read aloud.*

**1. 🐟 The main food of Southern Resident killer whales is…** Pretest #1

A. Seals | **B. Chinook salmon** | C. Krill

**2. 📉 The number of Southern Resident killer whales is…** Pretest #2

A. Increasing | B. Staying the same | **C. Decreasing**

**3. 👨‍👩‍👧‍👦 The three Southern Resident pods are…** A1.9 review

**A. J, K, and L** | B. A, B, and C | C. X, Y, and Z

**4. 🌿 In the Salish Sea food web, the producers are…** A1.9 review

A. Orcas | B. Seals | **C. Phytoplankton**

**5. 🕸️ A food web is…** A1.9 review

**A. Many food chains linked together** | B. One straight line of eating | C. A fishing net

**6. ☠️ Biomagnification means toxins…** A1.9 review

A. Disappear in the ocean | **B. Get more concentrated up the food chain** | C. Only hurt plants

**7. 🧈 PCBs are fat-soluble, so they…** A1.9 review

A. Wash out in urine | **B. Are stored in blubber** | C. Turn into salmon

**8. 🔊 Echolocation is…** A1.9 review

**A. Using clicks and echoes to find prey** | B. Seeing in the dark | C. Smelling fish

**9. 🚤 Extra boat noise makes orcas…** A1.9 review

A. Find food faster | B. Sleep more | **C. Work harder to find food**

**10. 💗 J32 died from…** A1.9 review

**A. Lack of food, toxins, and noise together** | B. Old age only | C. A shark

## Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

*Reworded from the gen ed "Orca preassessment + A1.9 Orca Unit Review". Score: __ / 10.*

🔑 *Key written by our team from her Orca preassessment and A1.9 review, please check. Her Orca Test (32 pts) is on paper and not in the export; this is a "From review" unit quiz.*